

Final Project Report | Group 19 | Mikael Granqvist, Joonas Salo, Janita Collin

This report describes our process of implementing a step-counting system utilizing the nRF52 development board in combination with the LSM6DSO sensor. The primary objective of the project was to implement pedometer functionality and transmit the resulting step count to a smartphone via a Bluetooth Low Energy (BLE) connection, using the nRF Connect and nRF Toolbox applications.

Our work began with finding a suitable method for implementing the step-counting algorithm. We determined that the pedometer functionality provided in the *lsm6dso_pedometer.c* file, available from the STMicroelectronics GitHub repository, was suitable for the project. We incorporated and tested this functionality within the existing *template_project* used in previous course exercises. The integration process proceeded as follows:

1. Importing variables:
 - a. `uint16_t steps`, `uint8_t tx_buffer[1000]`, `lsm6dso_emb_sens_t emb_sens`
2. Importing functions:
 - a. `lsm6dso_number_of_steps_get()`, `lsm6dso_steps_reset()`,
`lsm6dso_pedo_sens_set()`, `lsm6dso_embedded_sens_set()`
3. Testing functionality:
 - a. The pedometer routine was executed in the main loop, and the value of the steps variable was monitored via the watch window to verify correct incrementation.

Once the step-counting functionality had been successfully validated in the *template_project*, we transitioned to the *ble_app_uart* template, which served as the basis for the BLE communication. This template provides UART-over-BLE services, allowing us to establish a BLE connection.

Next, we added the required platform and register files for the LSM6DSO sensor. The *nrfx_twi.c* driver was also added into the project's driver folder, and the TWI interface was enabled via the CMSIS Configuration Wizard.

In the final stage, we integrated the *twi_init()*, *lsm6dso_setup()*, and *stepcounter()* functions, along with the necessary variable definitions and header file imports. The corresponding function calls were added to the main function, and *stepcounter()* was executed within the main loop. After completing these integrations, we tested the BLE functionality with a smartphone and successfully displayed the real-time step count on the mobile device. In the nRF Connect application the step count was displayed as the value of a TX Characteristic and in the nRF Toolbox application the step count was displayed in a message format.

